Scorecard & Credit Risk Calculations

MASTER in AI, USJ, ESIB, Mansourieh

AI in Fintech

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1- Develop a scoring model

TrainingData =
DataProjScoreCard(:,{'Age','ResidentialStatus','EmploymentStatus','Income','Default'});
sc = creditscorecard(TrainingData, ...
    'ResponseVar','Default', ...
    'GoodLabel',0);
disp(sc);

Score Card

sc ×

1×1 creditscorecard

Property	Value	Size	Class
GoodLabel	0	1×1	double
ResponseVar	'Default'	1×7	char
WeightsVar	''	0×0	char
VarNames	1×5 cell	1×5	cell
NumericPredictors	1×2 cell	1×2	cell
CategoricalPredictors	1×2 cell	1×2	cell
BinMissingData	0	1×1	logical
IDVar	''	0×0	char
PredictorVars	1×4 cell	1×4	cell
Data	500×5 table	500×5	table

	1	2	3	4	5
1	'Age'	'ResidentialStatus'	'EmploymentStatus'	'Income'	'Default'

	1	2
1	'Age'	'Income'

	1	2
1	'ResidentialStatus'	'EmploymentStatus'

	1	2	3	4
1	'Age'	'ResidentialStatus'	'EmploymentStatus'	'Income'

sc × sc.Data ×

500×5 table

	1 Age	2 ResidentialStatus	3 EmploymentStatus	4 Income	5 Default
1	25	renter	Other	21000	1
2	63	HomeOwner	Employed	53000	0
3	30	renter	Other	30000	1
4	52	HomeOwner	Employed	54000	0
5	70	HomeOwner	Employed	54000	0
6	67	HomeOwner	Employed	49000	0
7	36	HomeOwner	Other	33000	1
8	52	renter	Employed	52000	0
9	35	renter	Other	23000	1
10	37	renter	Other	26000	1

Command window view:

Command Window

```
  creditscorecard with properties:

              GoodLabel: 0
            ResponseVar: 'Default'
             WeightsVar: ''
               VarNames: {'Age'  'ResidentialStatus'  'EmploymentStatus'  'Income'  'Default'}
      NumericPredictors: {'Age'  'Income'}
  CategoricalPredictors: {'ResidentialStatus'  'EmploymentStatus'}
         BinMissingData: 0
                  IDVar: ''
          PredictorVars: {'Age'  'ResidentialStatus'  'EmploymentStatus'  'Income'}
                   Data: [500×5 table]
```

2- Bin the data and show the WOE

sc = autobinning(sc);

2-1- Age binning

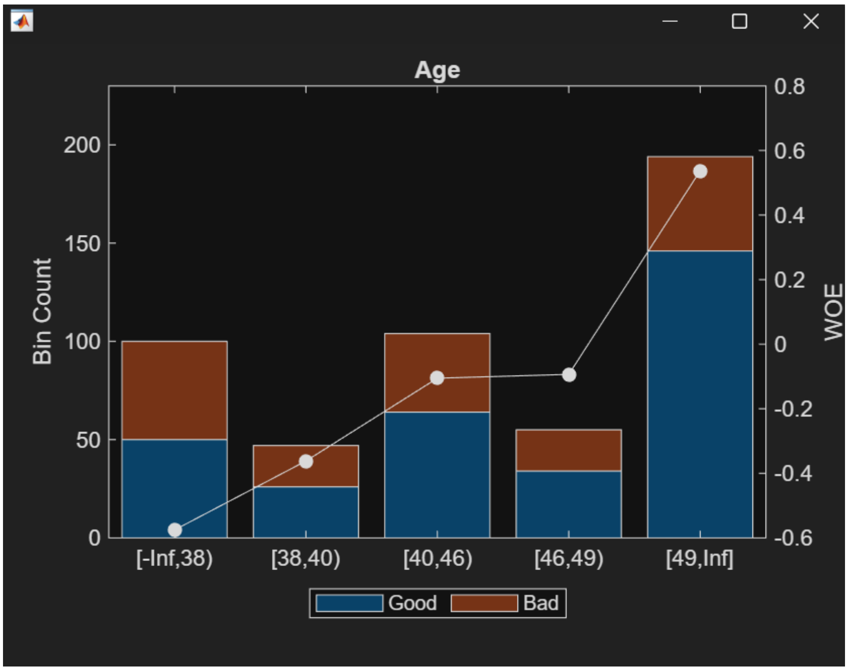
[bi1, cp1] = **bininfo**(sc, 'Age');
disp('Binning Information for Age');
disp(bi1);
plotbins(sc, 'Age');

bi1 — 6×6 table

	1 Bin	2 Good	3 Bad	4 Odds	5 WOE	6 InfoValue
1	'[-Inf,38)'	50	50	1	-0.5754	0.0699
2	'[38,40)'	26	21	1.2381	-0.3618	0.0128
3	'[40,46)'	64	40	1.6000	-0.1054	0.0023
4	'[46,49)'	34	21	1.6190	-0.0935	9.7423e-04
5	'[49,Inf]'	146	48	3.0417	0.5370	0.1018
6	'Totals'	320	180	1.7778	NaN	0.1879

cp1 — 4×1 double

	1
1	38
2	40
3	46
4	49



2-2- Income binning

```
[bi2, cp2] = bininfo(sc, 'Income');
disp('Binning Information for Income');
disp(bi2);
plotbins(sc, 'Income');
```

bi2 × cp2 ×

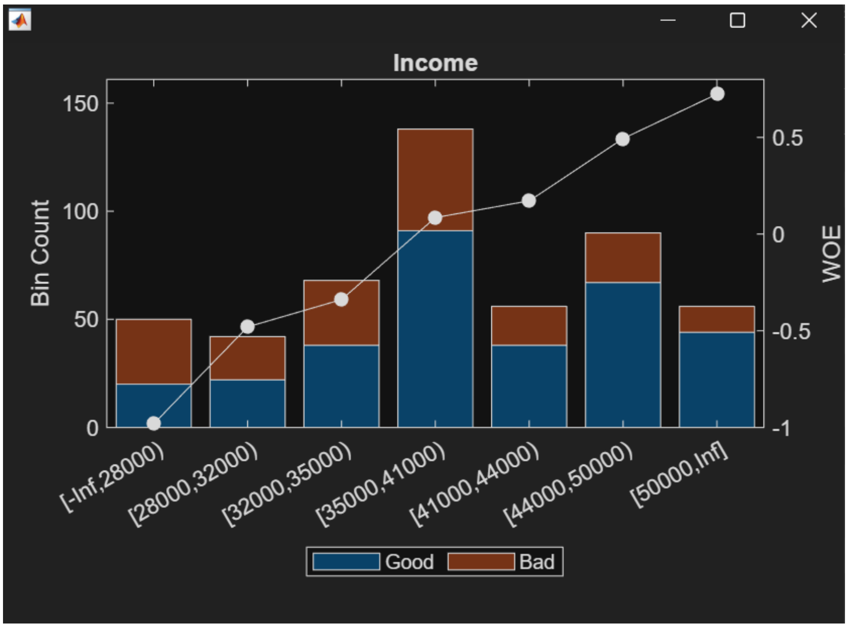
8×6 table

	1 Bin	2 Good	3 Bad	4 Odds	5 WOE	6 InfoValue
1	'[-Inf,28000)'	20	30	0.6667	-0.9808	0.1022
2	'[28000,32000)'	22	20	1.1000	-0.4801	0.0203
3	'[32000,35000)'	38	30	1.2667	-0.3390	0.0162
4	'[35000,41000)'	91	47	1.9362	0.0853	0.0020
5	'[41000,44000)'	38	18	2.1111	0.1719	0.0032
6	'[44000,50000)'	67	23	2.9130	0.4938	0.0403
7	'[50000,Inf]'	44	12	3.6667	0.7239	0.0513
8	'Totals'	320	180	1.7778	NaN	0.2355

bi2 × cp2 ×

6×1 double

	1
1	28000
2	32000
3	35000
4	41000
5	44000
6	50000



2-3- Residential Status binning

```
[bi3, cp3] = bininfo(sc, 'ResidentialStatus');
disp('Binning Information for ResidentialStatus');
disp(bi3);
plotbins(sc, 'ResidentialStatus');
```

bi3 × cp3 ×

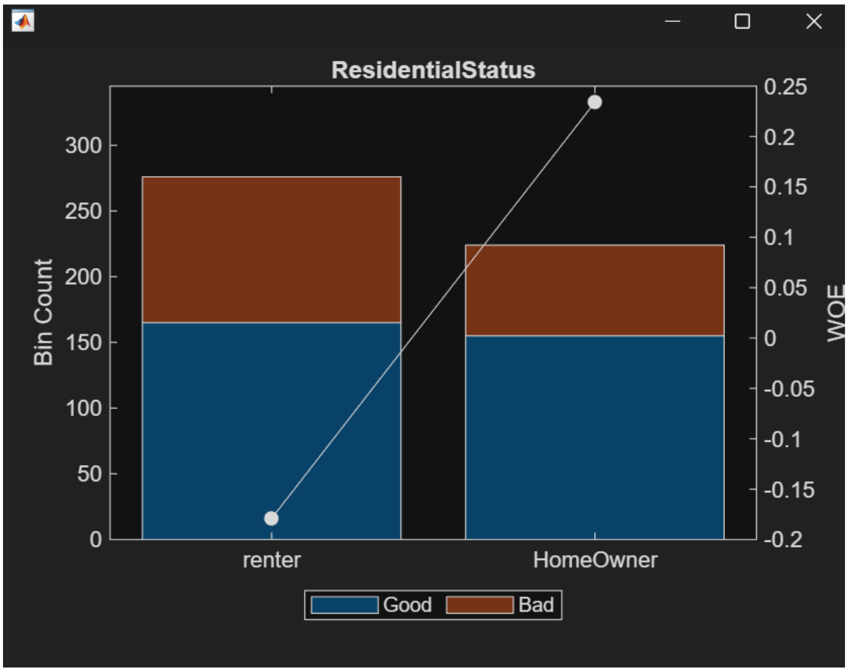
3×6 table

	1 Bin	2 Good	3 Bad	4 Odds	5 WOE	6 InfoValue
1	'renter'	165	111	1.4865	-0.1789	0.0181
2	'HomeOwner'	155	69	2.2464	0.2340	0.0236
3	'Totals'	320	180	1.7778	NaN	0.0417

bi3 × cp3 ×

2×2 table

	1 Category	2 BinNumber
1	'renter'	1
2	'HomeOwner'	2



2-4- Employment Status binning

```
[bi4, cp4] = bininfo(sc, 'EmploymentStatus');
disp('Binning Information for EmploymentStatus');
disp(bi4);
plotbins(sc, 'EmploymentStatus');
```

cp4 × bi4 ×

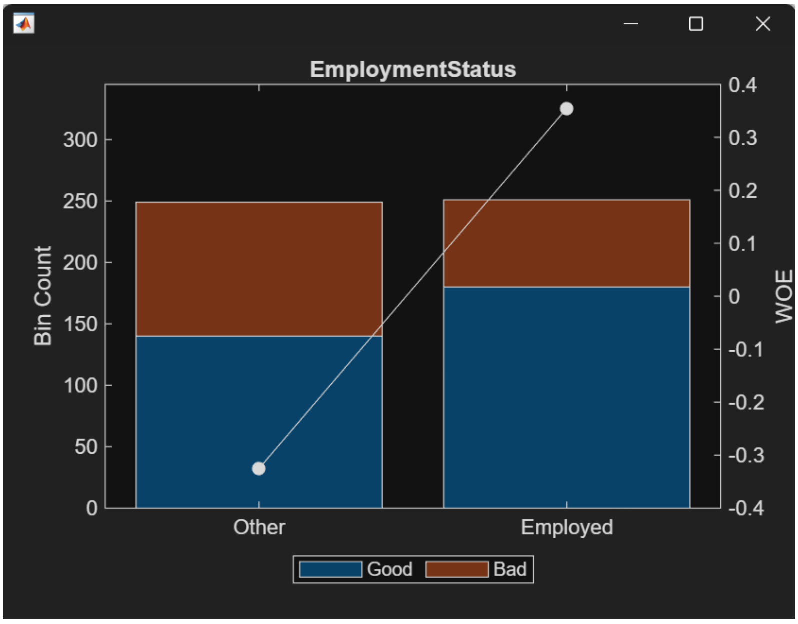
3×6 table

	1 Bin	2 Good	3 Bad	4 Odds	5 WOE	6 InfoValue
1	'Other'	140	109	1.2844	-0.3251	0.0546
2	'Employed'	180	71	2.5352	0.3549	0.0596
3	'Totals'	320	180	1.7778	NaN	0.1143

cp4 × bi4 ×

2×2 table

	1 Category	2 BinNumber
1	'Other'	1
2	'Employed'	2



3- Fit a logistic regression for the model

sc = fitmodel(sc);
disp(sc);

Command window view:

```
Command Window

1. Adding Income, Deviance = 626.6132, Chi2Stat = 26.80497, PValue = 2.250573e-07
2. Adding EmploymentStatus, Deviance = 616.8266, Chi2Stat = 9.786585, PValue = 0.001757897
3. Adding Age, Deviance = 611.6613, Chi2Stat = 5.165336, PValue = 0.023042
4. Adding ResidentialStatus, Deviance = 607.4818, Chi2Stat = 4.17948, PValue = 0.04091624

Generalized linear regression model:
    logit(Default) ~ 1 + Age + ResidentialStatus + EmploymentStatus + Income
    Distribution = Binomial

Estimated Coefficients:
                         Estimate       SE       tStat       pValue  
                         ________    ________    ______    __________

    (Intercept)          0.57229     0.097901    5.8456    5.0482e-09
    Age                  0.62054      0.27659    2.2435      0.024863
    ResidentialStatus    0.97467      0.47898    2.0349      0.041862
    EmploymentStatus     0.95864      0.29081    3.2965    0.00097901
    Income               0.61637      0.24568    2.5088      0.012115


500 observations, 495 error degrees of freedom
Dispersion: 1
Chi^2-statistic vs. constant model: 45.9, p-value = 2.54e-09
```

4- Display the unscaled points by using the function (display points)

points = **displaypoints**(sc);
disp('Unscaled Points');
disp(points);

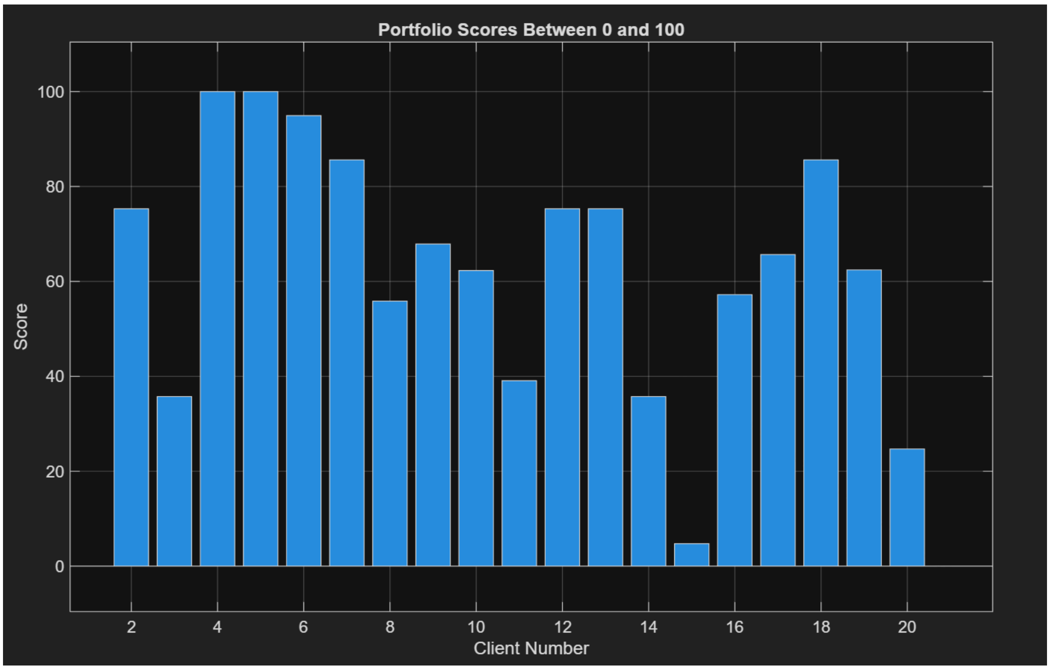
```
Unscaled Points
         Predictors                  Bin             Points
    _____________________    ________________    _________

    {'Age'              }    {'[-Inf,38)'    }    -0.21396
    {'Age'              }    {'[38,40)'      }   -0.081434
    {'Age'              }    {'[40,46)'      }    0.077691
    {'Age'              }    {'[46,49)'      }    0.085035
    {'Age'              }    {'[49,Inf]'     }     0.47633
    {'Age'              }    {'<missing>'    }         NaN
    {'ResidentialStatus'}    {'renter'       }   -0.031346
    {'ResidentialStatus'}    {'HomeOwner'    }      0.3711
    {'ResidentialStatus'}    {'<missing>'    }         NaN
    {'EmploymentStatus' }    {'Other'        }    -0.16855
    {'EmploymentStatus' }    {'Employed'     }     0.48331
    {'EmploymentStatus' }    {'<missing>'    }         NaN
    {'Income'           }    {'[-Inf,28000)' }    -0.46148
    {'Income'           }    {'[28000,32000)'}    -0.15282
    {'Income'           }    {'[32000,35000)'}   -0.065862
    {'Income'           }    {'[35000,41000)'}     0.19568
    {'Income'           }    {'[41000,44000)'}     0.24899
    {'Income'           }    {'[44000,50000)'}     0.44746
    {'Income'           }    {'[50000,Inf]'  }     0.58927
    {'Income'           }    {'<missing>'    }         NaN
```

5- Mapping Portfolio scores between 0 and 100

sc = formatpoints(sc, 'WorstAndBestScores', [0 100]);
Scores = **score**(sc, ActualPortfolioData);
disp('Portfolio Scores');
disp(Scores);

```
Portfolio Scores
   -0.0000
   75.3057
   35.7363
  100.0000
  100.0000
   94.9267
   85.6030
   55.8354
   67.8739
   62.2836
   39.0913
   75.3057
   75.3057
   35.7363
    4.7411
   57.2103
   65.6499
   85.6030
   62.4199
   24.6943
```



NB: Client 1 is not missing in the histogram, it's just that his mapped portfolio score is equal to 0.

6- Plot the ROC curve using the Historical Data

```
[Stats, T] = validatemodel(sc, 'Plot', 'ROC');
disp('Validation Statistics');
disp(Stats);
```

Command Window

Validation Statistics

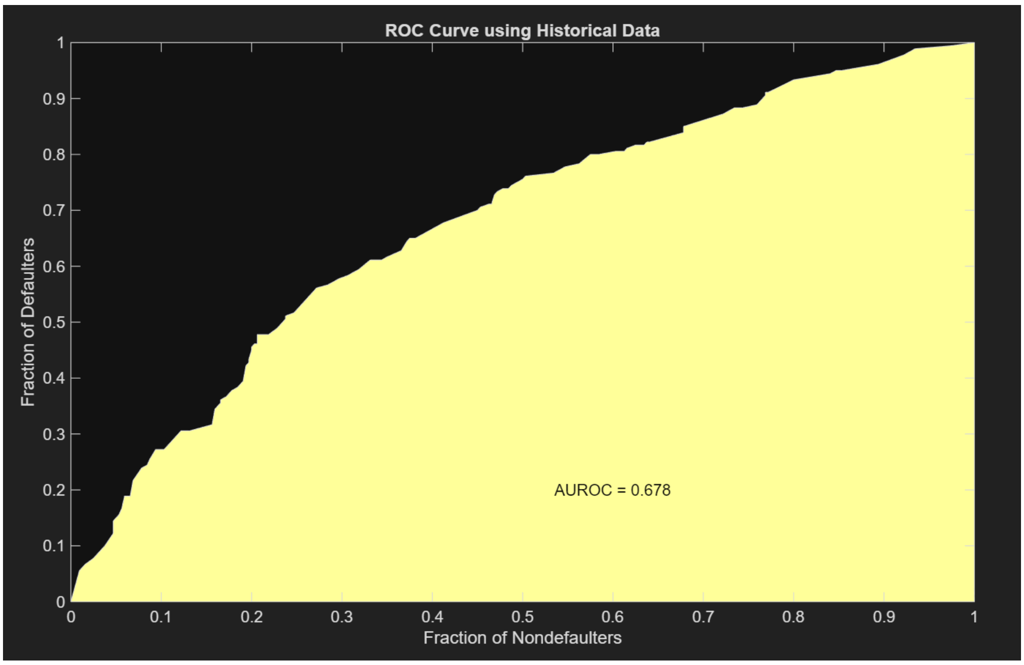
Measure	Value
{'Accuracy Ratio' }	0.35507
{'Area under ROC curve'}	0.67753
{'KS statistic' }	0.28924
{'KS score' }	48.203

```
disp('ROC Table');
disp(T);
```

Command Window

ROC Table

Scores	ProbDefault	TrueBads	FalseBads	TrueGoods	FalseGoods	Sensitivity	FalseAlarm	PctObs
-3.5527e-15	0.70586	10	3	317	170	0.055556	0.009375	0.026
4.7411	0.67761	12	5	315	168	0.066667	0.015625	0.034
10.434	0.64192	14	8	312	166	0.077778	0.025	0.044
11.042	0.638	16	10	310	164	0.088889	0.03125	0.052
14.153	0.61768	18	12	308	162	0.1	0.0375	0.06
14.397	0.61607	22	15	305	158	0.12222	0.046875	0.074
15.783	0.60686	26	15	305	154	0.14444	0.046875	0.082
18.894	0.58594	28	17	303	152	0.15556	0.053125	0.09
19.138	0.58428	30	18	302	150	0.16667	0.05625	0.096
21.476	0.56833	34	19	301	146	0.18889	0.059375	0.106
21.738	0.56652	34	21	299	146	0.18889	0.065625	0.11
23.319	0.55564	39	22	298	141	0.21667	0.06875	0.122
23.509	0.55433	43	25	295	137	0.23889	0.078125	0.136
24.586	0.54688	44	27	293	136	0.24444	0.084375	0.142
24.831	0.54519	46	28	292	134	0.25556	0.0875	0.148
24.849	0.54506	49	30	290	131	0.27222	0.09375	0.158
25.093	0.54337	49	32	288	131	0.27222	0.1	0.162
25.416	0.54112	49	33	287	131	0.27222	0.10312	0.164
25.439	0.54097	51	35	285	129	0.28333	0.10938	0.172
28.061	0.52272	52	36	284	128	0.28889	0.1125	0.176
28.25	0.5214	53	37	283	127	0.29444	0.11563	0.18



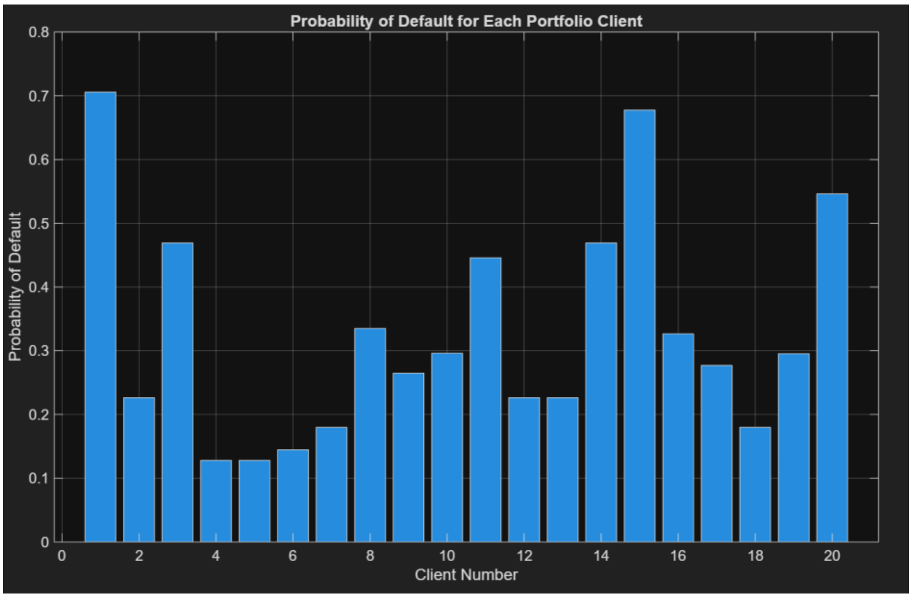
The model is about **67.8% able** to correctly distinguish between a good client and a defaulting client based on their scores.

7- Probability of default

PDs = probdefault(sc, ActualPortfolioData);
disp('Probability of Default for Portfolio Clients');
disp(PDs);

Command Window

```
Probability of Default for Portfolio Clients
    0.7059
    0.2262
    0.4691
    0.1279
    0.1279
    0.1445
    0.1798
    0.3350
    0.2646
    0.2961
    0.4459
    0.2262
    0.2262
    0.4691
    0.6776
    0.3265
    0.2769
    0.1798
    0.2954
    0.5461
```



How is the probability of default mathematically computed?

Let S be the unscaled score,
b0,b1,b2,b3,b4 be the logistic regression coefficients,
and WOE_Age is the Weight of Evidence value of the age bin where the client belongs.

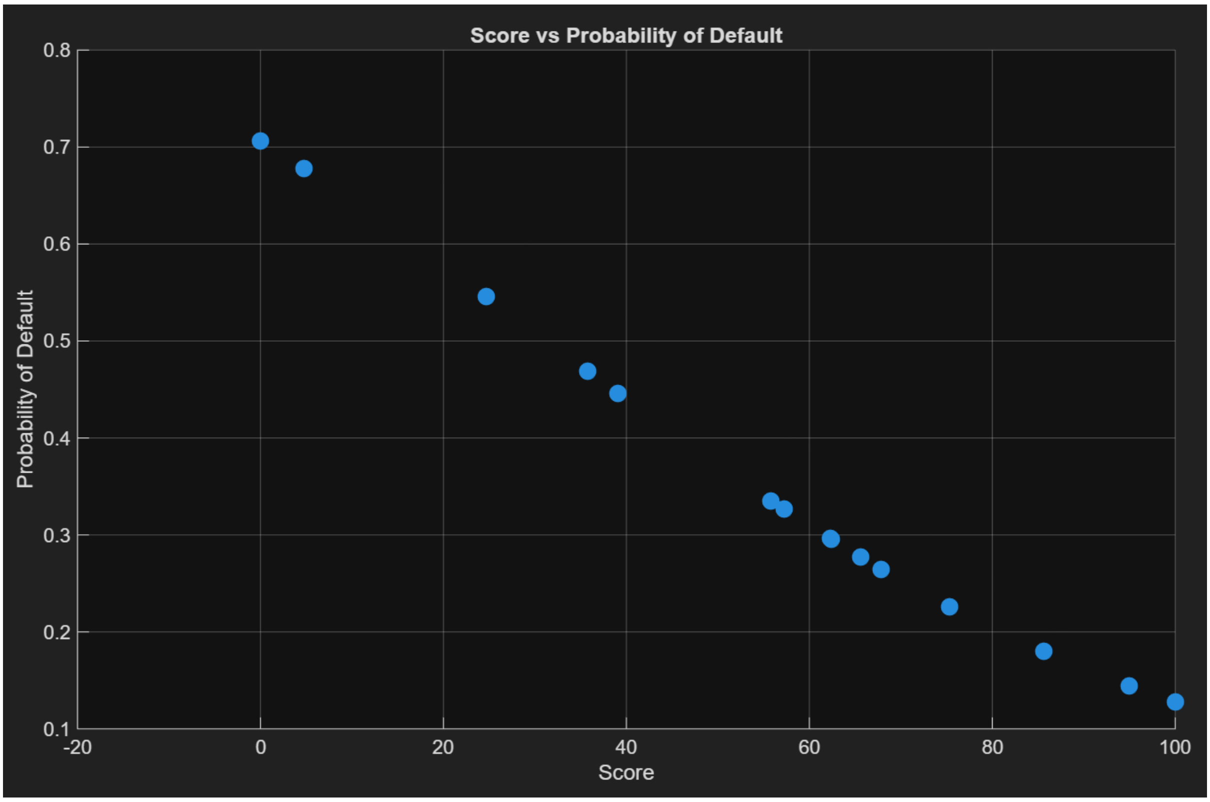
S = b0
+ b1*WOE_Age
+ b2*WOE_ResidentialStatus
+ b3*WOE_EmploymentStatus
+ b4*WOE_Income

Therefore, to calculate the survival probability we simply compute

Probability of survival = $\frac{1}{1+e^{-S}}$

Consequentially, Probability of default = (1 - Probability of survival)

As we can see in the following plot, that the higher the probability of default, the lower is the score of the client.



8- Find the optimal score and calculate the expected loss of the optimal selected portfolio given a recovery rate of 60%

8-1- Optimal score

Find the optimal score using the Optimal cut-off score (OCO)
OCO is simply equal to the KS score from validatemodel function that we ran earlier.

```
Command Window
Validation Statistics
            Measure                 Value
    _______________________        _______

    {'Accuracy Ratio'       }      0.35507
    {'Area under ROC curve' }      0.67753
    {'KS statistic'         }      0.28924
    {'KS score'             }       48.203
```

Another method to find the optimal score

Compute the sum of **TrueGoods + TrueBads** for each possible score threshold, then choose the score with the maximum sum of correct predictions.

We end up with a threshold score of **43.21** for a maximum sum of 340.

optimalScore = Stats.Value(strcmp(Stats.Measure, 'KS score'));
disp('Optimal Cut-Off Score');
disp(optimalScore);

```
Command Window
Optimal Cut-Off Score
   48.2032
```

If Score >= **48.20** ➔ Accept client
If Score < **48.20** ➔ Reject client

8-2- Select the good clients

Since higher score means lower risk, we select clients with a Score >= **optimalScore**

selectedIdx = Scores >= optimalScore;
disp('Selected Clients Index');
disp(selectedIdx);
% Keep only accepted clients
selectedClients = ActualPortfolioData(selectedIdx, :);
selectedClients.SelectedIdx = selectedIdx(selectedIdx);
selectedClients.ModelPD = PDs(selectedIdx);
selectedClients = movevars(selectedClients, 'SelectedIdx', 'Before', 'ID');
disp('Selected Clients Decision Table');
disp(selectedClients);

```
Selected Clients Index
   0
   1
   0
   1
   1
   1
   1
   1
   1
   1
   0
   1
   1
   0
   0
   1
   1
   1
   1
   0
```

Clients Index:
1 ➔ Accepted
0 ➔ Rejected

This table shows only the accepted clients along with their corresponding predicted probability of default.

Selected Clients Decision Table

SelectedIdx	ID	Age	ResidentialStatus	EmploymentStatus	Income	PD	LGD	NA	EL	ModelPD
true	2	22	HomeOwner	Employed	53000	0.1279	0.4	1e+05	5116	0.22623
true	4	50	HomeOwner	Employed	54000	0.1279	0.4	1e+05	5116	0.12786
true	5	60	HomeOwner	Employed	54000	0.1279	0.4	1e+05	5116	0.12786
true	6	77	HomeOwner	Employed	49000	0.1445	0.4	1e+05	5780	0.14453
true	7	63	renter	Employed	51300	0.5193	0.4	1e+05	20772	0.17982
true	8	29	renter	Employed	44000	0.1798	0.4	1e+05	7192	0.33505
true	9	38	HomeOwner	Employed	41000	0.7059	0.4	1e+05	28236	0.26464
true	10	57	renter	Other	68700	0.7059	0.4	1e+05	28236	0.29615
true	12	25	HomeOwner	Employed	53800	0.2448	0.4	1e+05	9792	0.22623
true	13	32	HomeOwner	Employed	66500	0.2448	0.4	1e+05	9792	0.22623
true	16	63	renter	Other	45400	0.3024	0.4	1e+05	12096	0.32654
true	17	39	renter	Employed	68500	0.2446	0.4	1e+05	9784	0.27692
true	18	68	renter	Employed	64700	0.5556	0.4	1e+05	22224	0.17982
true	19	40	HomeOwner	Other	66000	0.1279	0.4	1e+05	5116	0.29536

8-3- Compute the expected loss for the selected clients

Set the constants given:

loanAmount = 100000;

recoveryRate = 0.60;

LGD = 1 - recoveryRate;

LGD is Loss Given Default so here it is equal to 0.40

Then select the selected or accepted clients:

selectedPDs = PDs(selectedIdx);

Computed the expected loss for the selected clients (in $):

expectedLosses = selectedPDs * loanAmount * LGD;

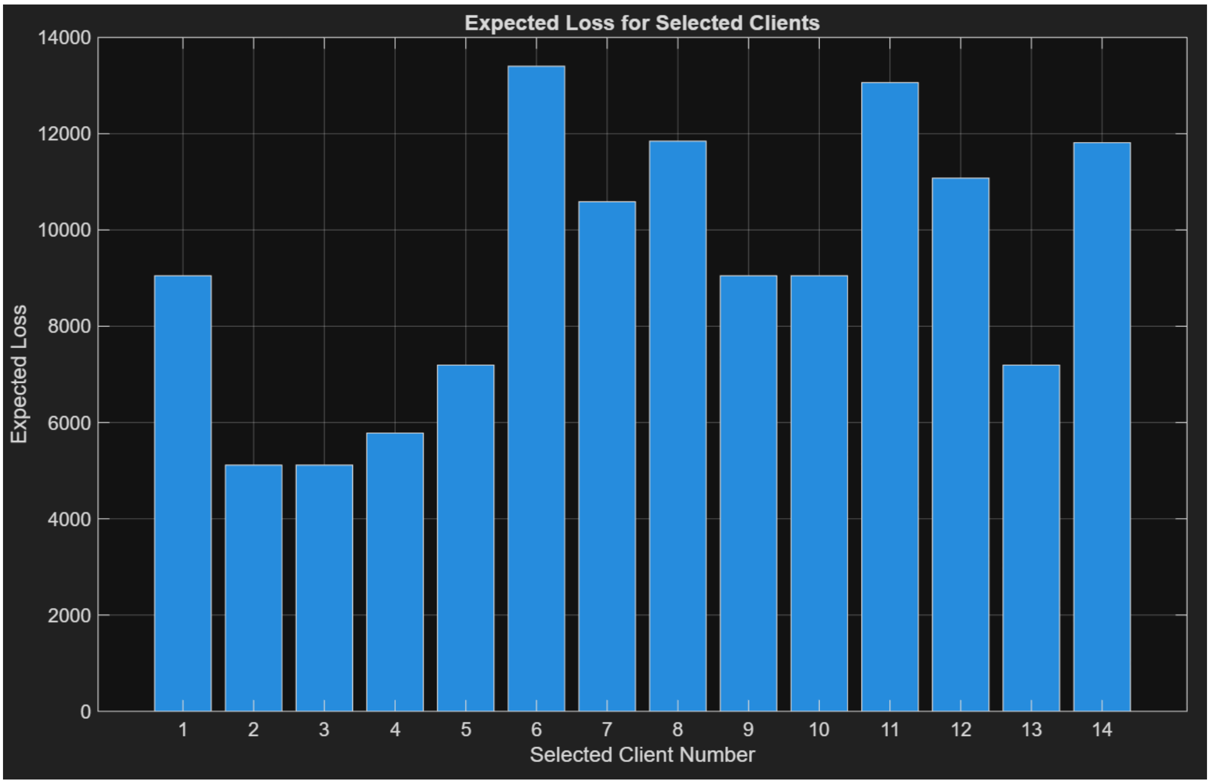
Expected Loss Table

ClientID	ExpectedLoss
2	9049.3
4	5114.4
5	5114.4
6	5781.1
7	7192.9
8	13402
9	10586
10	11846
12	9049.3
13	9049.3
16	13062
17	11077
18	7192.9
19	11814

Total expected loss (value in $):

```
totalExpectedLoss = sum(expectedLosses);
fprintf('Total Expected Loss: %.6f\n', totalExpectedLoss);
```

```
Total Expected Loss: 129329.627181
```



9- Price the minimum interest rate the bank should earn from every client in the optimal selected portfolio

minInterestRates = (selectedPDs * LGD) ./ (1 - selectedPDs);

Minimum Interest Rate Table

ClientID	MinimumInterestRate
2	0.11695
4	0.058643
5	0.058643
6	0.067578
7	0.087699
8	0.20155
9	0.14395
10	0.1683
12	0.11695
13	0.11695
16	0.19395
17	0.15319
18	0.087699
19	0.16766

The overall summary of the accepted clients

```
selectedClients.PD = selectedPDs;
selectedClients.ExpectedLoss = expectedLosses;
selectedClients.MinInterestRate = minInterestRates;
disp('Final Selected Clients');
disp(selectedClients);
```

Final Selected Clients

SelectedIdx	ID	Age	ResidentialStatus	EmploymentStatus	Income	PD	LGD	NA	EL	ModelPD	ExpectedLoss	MinInterestRate
true	2	22	HomeOwner	Employed	53000	0.22623	0.4	1e+05	5116	0.22623	9049.3	0.11695
true	4	50	HomeOwner	Employed	54000	0.12786	0.4	1e+05	5116	0.12786	5114.4	0.058643
true	5	60	HomeOwner	Employed	54000	0.12786	0.4	1e+05	5116	0.12786	5114.4	0.058643
true	6	77	HomeOwner	Employed	49000	0.14453	0.4	1e+05	5780	0.14453	5781.1	0.067578
true	7	63	renter	Employed	51300	0.17982	0.4	1e+05	20772	0.17982	7192.9	0.087699
true	8	29	renter	Employed	44000	0.33505	0.4	1e+05	7192	0.33505	13402	0.20155
true	9	38	HomeOwner	Employed	41000	0.26464	0.4	1e+05	28236	0.26464	10586	0.14395
true	10	57	renter	Other	68700	0.29615	0.4	1e+05	28236	0.29615	11846	0.1683
true	12	25	HomeOwner	Employed	53800	0.22623	0.4	1e+05	9792	0.22623	9049.3	0.11695
true	13	32	HomeOwner	Employed	66500	0.22623	0.4	1e+05	9792	0.22623	9049.3	0.11695
true	16	63	renter	Other	45400	0.32654	0.4	1e+05	12096	0.32654	13062	0.19395
true	17	39	renter	Employed	68500	0.27692	0.4	1e+05	9784	0.27692	11077	0.15319
true	18	68	renter	Employed	64700	0.17982	0.4	1e+05	22224	0.17982	7192.9	0.087699
true	19	40	HomeOwner	Other	66000	0.29536	0.4	1e+05	5116	0.29536	11814	0.16766